7.3. ANALYSIS OF COUPLED CIRCUIT

Consider the circuit shown in Fig. 7.4, where two windings are wound on the common core. The self inductances of windings are L_1 and L_2 while M is the mutual inductance between them.

The sign of M is decided by the winding sense. If the current in two coils producing flux in such a way that two fluxes are supporting

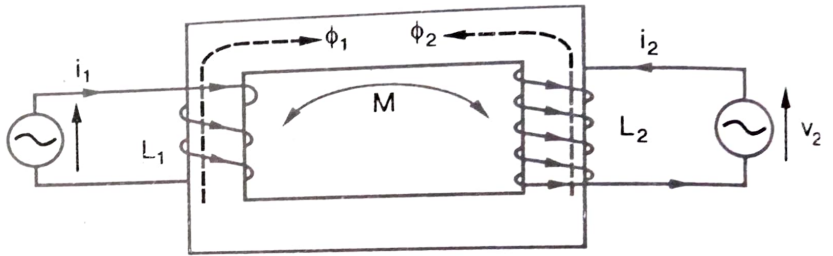


Fig. 7.4.

each other then sign of M is taken positive. If the fluxes are opposing each other then M is considered negative. In the Fig. 7.4, ϕ_1 and ϕ_2 are opposing each other so M will be negative. The KVL equations for the given circuit are —

loop 1 :
$$v_1 = L_1 \frac{di_1}{dt} - M \frac{di_2}{dt} \quad \ldots(7.25)$$

loop 2 :
$$v_2 = L_2 \frac{di_2}{dt} - M \frac{di_1}{dt} \quad \ldots(7.26)$$

Now we change the winding sense as shown in Fig. 7.5.

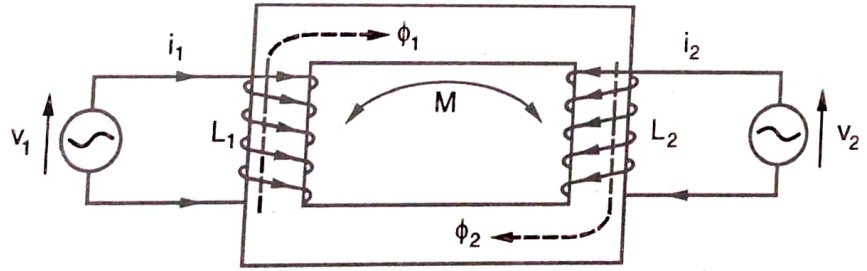


Fig. 7.5.

Now ϕ_1 and ϕ_2 supporting each other so M will be positive, the KVL equations are :

loop 1 :
$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt} \quad \ldots(7.27)$$

loop 2 :
$$v_2 = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt} \quad \ldots(7.28)$$

7.4. DOT CONVENSION FOR COUPLED CIRCUIT

Placement of dots on the coil, decides the sign of *M*. To place the dot on any coil coupled with other coil, we flow the current in one coil of pair and assign the dot where the current enters the coil. Now the fluxes of two coils are opposing each other (*M* negative) then dot in the second coil will be place where current is leaving the coil. In Fig. 7.4, we place the dots. (Fig. 7.6).

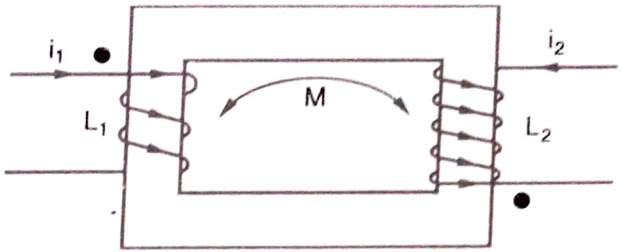


Fig. 7.6. Dotted Circuit for Fig. 7.4.

The dotted equivalent circuit is shown in Fig. 7.7.

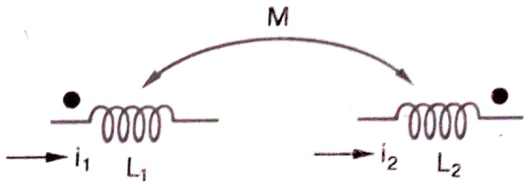


Fig. 7.7. Dotted Equivalent Circuit

Now consider the case of Fig. 7.5, where two fluxes are supporting each other. Here dots are placed in both the coils where currents are entering to the coils.

The dotted equivalent circuit is shown in Fig 7.9, M is positive.

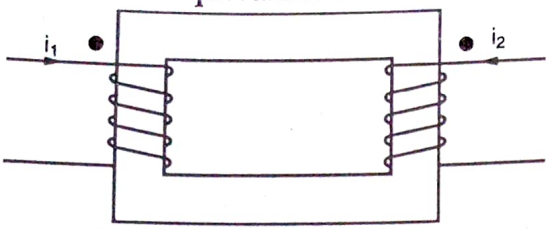


Fig. 7.8. Dotted Circuit for Fig. 7.5.

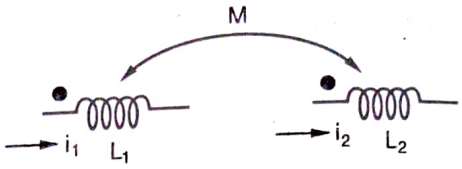


Fig. 7.9. Dotted Equivalent Circuit.

So the Dot convension (dot rule) is :

(*i*) Currents in both the coils entering at dots.

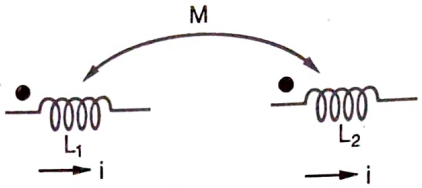


Fig. 7.10.

M will have the same sign as that of L_1 and L_2 *i.e.*, M will be positive.

(*ii*) Currents in both the coils leaving at dots.

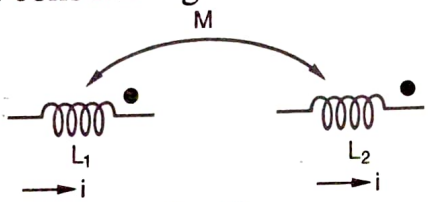


Fig. 7.11.

This case is same as case (*i*) *i.e.*, M will be positive.

(*iii*) Current in one coil entering at dot while in other it is leaving at dot.

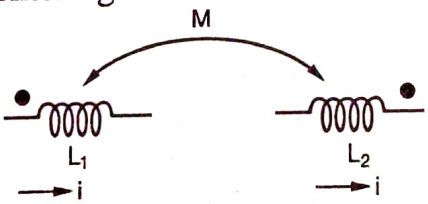


Fig. 7.12.

M will have the sign opposite to L_1 and L_2 *i.e.*, M will be negative.

(*iv*) This case is same as case (*iii*) *i.e.*, M will be negative. (Fig. 7.13.)

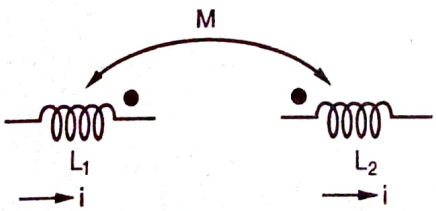


Fig. 7.13.

Maxwells loop matrix for dotted equivalent circuit :

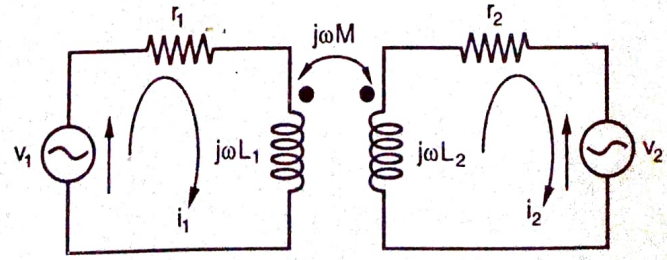


Fig. 7.14.

For loop 1 i_1 is entering to the coil at dot, while in loop 2, i_2 is leaving the coil at dot so by dot convension M will be negative. So the KVL equation in matrix form can be directly written as :

$$\begin{bmatrix} r_1 + j\omega L_1 & -j\omega M \\ -j\omega M & r_2 + j\omega L_2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ -v_2 \end{bmatrix} \quad ...(7.29)$$

Now i_1 and i_2 can be calculated by Cramer's rule as explained in chapter 6.

7.5. CONDUCTIVELY COUPLED CIRCUIT

In Fig. 7.14, two coils are mutually coupled while they are electrically isolated. The conductively coupled circuit for Fig. 7.14 can be drawn as shown in Fig. 7.15.

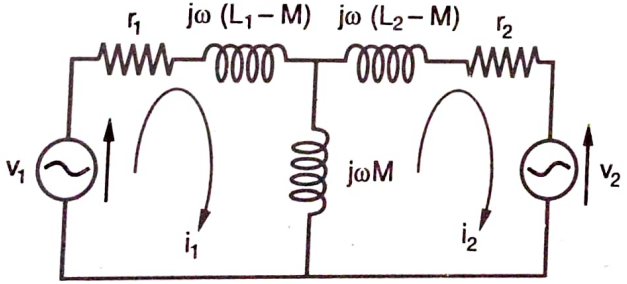


Fig. 7.15. Conductivity Coupled Circuit.

If we write the KVL equation for the above circuit then we get the same result as we got for the circuit shown in Fig. 7.14 (*i.e.*, equation 7.29).

Procedure for drawing the conductively coupled circuit. First of all connect the mutual impedance in the common branch of two loops, then to nullify its effect on self impedances of two loop it must be subtracted from the self impedances of each loop. This method of analysis does not always give the physically realizable equivalent circuit. This is true when $M > L_1$ or $M > L_2$.

equivalent circuit.

7.6. SERIES AND PARALLEL COMBINATION OF MUTUALLY COUPLED CIRCUITS

(a) Series combination.

(i) **Direct series combination.** (fluxes of both the coils are adding each other) : Both the coils are carrying the same current i and two emfs are induced in both the coils, one is due to its self inductance and other is due to mutual inductance.

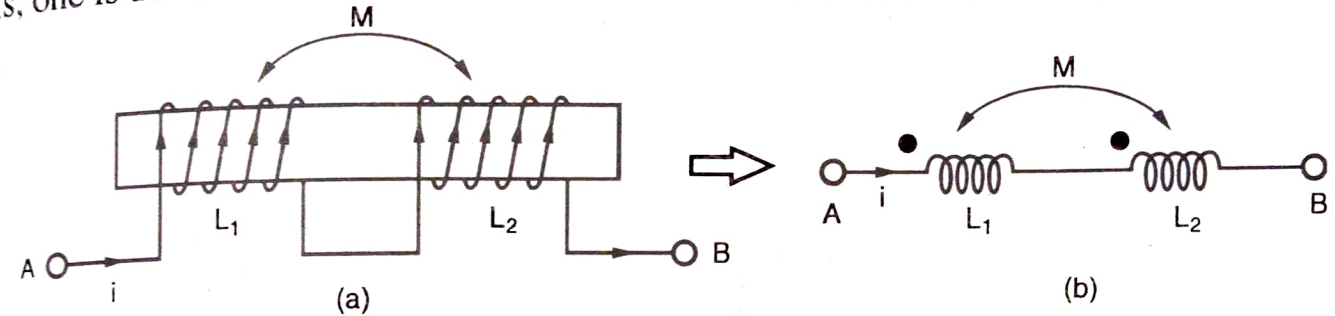


Fig. 7.16. Series Combination (M Positive)

Total emf induced in coil one :

$$e_1 = L_1 \frac{di}{dt} + M \frac{di}{dt} \qquad ...(7.30)$$

(here M is positive by dot convension) similarly total emf induced in coil two :

$$e_2 = L_2 \frac{di}{dt} + M \frac{di}{dt} \qquad ...(7.31)$$

So the total emf induced between terminals A and B :

$$e = e_1 + e_2 = (L_1 + L_2 + 2M)\frac{di}{dt} \qquad ...(7.32)$$

But $$e = L_{eq}\frac{di}{dt}$$

So $$L_{eq} = L_1 + L_2 + 2M \qquad ...(7.33)$$

(*ii*) Series opposition (fluxes of two coils are opposing each other).

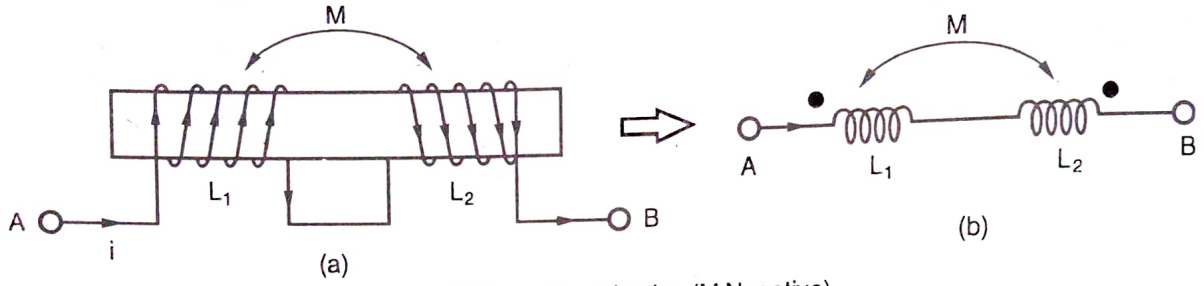


Fig. 7.17. Series Combination (M Negative)

Here M will be negative by dot convension. So total emf induced in coil 1 is

$$e_1 = L_1\frac{di}{dt} - M\frac{di}{dt} \qquad ...(7.34)$$

and emf induced in coil 2 is

$$e_2 = L_2\frac{di}{dt} - M\frac{di}{dt} \qquad ...(7.35)$$

$$\therefore \quad e = e_1 + e_2 = (L_1 + L_2 - 2M)\frac{di}{dt} \qquad ...(7.36)$$

So the equivalent inductance of this combination is

$$L_{eq} = L_1 + L_2 - 2M \qquad ...(7.37)$$

(*b*) **Parallel combination.** (*i*) Direct parallel (M is positive).

Since i_1 and i_2 both are entering to coils at dot so M will be positive. Both coils are parallel so voltage across them will be same.

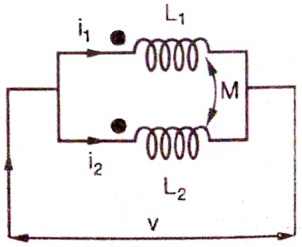


Fig. 7.18. Parallel

$$L_1\frac{di_1}{dt} + M\frac{di_2}{dt} = v \qquad ...(7.38)$$

and $$L_2\frac{di_2}{dt} + M\frac{di_1}{dt} = v \qquad ...(7.39)$$

by solving these two equation for $\frac{di_1}{dt}$ and $\frac{di_2}{dt}$, we get :

$$\frac{di_1}{dt} = \frac{(M - L_2)\,v}{M^2 - L_1 L_2} = \left(\frac{L_2 - M}{L_1 L_2 - M^2}\right)v \qquad ...(7.40)$$

and $$\frac{di_2}{dt} = \left(\frac{M - L_1}{M^2 - L_1 L_2}\right)v = \left(\frac{L_1 - M}{L_1 L_2 - M^2}\right)v \qquad ...(7.41)$$

by KCL we can write

$$i = i_1 + i_2 \qquad ...(7.42)$$

$$\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt} \qquad ...(7.43)$$

Put the values of $\frac{di_1}{dt}$ and $\frac{di_2}{dt}$ in equation 7.43

$$\frac{di}{dt} = \left[\frac{L_2 - M}{L_1 L_2 - M^2} + \frac{L_1 - M}{L_1 L_2 - M^2} \right] v = \left[\frac{L_1 + L_2 - 2M}{L_1 L_2 - M^2} \right] v \qquad ...(7.44)$$

$$v = \left[\frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} \right] \frac{di}{dt} \qquad ...(7.45)$$

but $$v = L_{eq} \frac{di}{dt}$$

so $$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} \qquad ...(7.46)$$

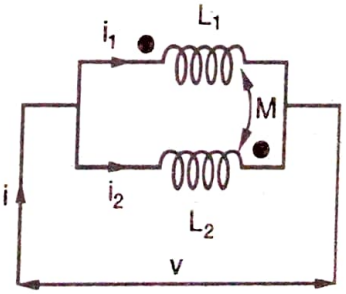


Fig. 7.19. Parallel Combination (M Negative)

(*ii*) Parallel opposition [*M* is negative]

Here i_1 is entering to coil at dot while i_2 is leaving the coil at dot so here M is negative. For this case the equivalent inductance is given by

$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M} \qquad ...(7.47)$$